

Style-Aware Data Augmentation for Deep Learning on Symbolic Music

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ABSTRACT

This study proposes a style-aware data augmentation framework that combines rule-based design with statistical constraints, applied to small-scale, highly constrained Jiangnan symbolic music generation tasks. By leveraging YNote representation, fixed rhythmic frameworks, and Markov-style local transition statistics, we systematically expand the training data while maintaining musical structural plausibility. Using the augmented data, we fine-tune a GPT-2 model to analyze how different training data scales affect generation behavior. Experimental results show that BLEU-based reference-overlap metrics exhibit only minor fluctuations across different training scales and are insufficient to directly reflect style improvement. In contrast, KL divergence and bigram statistics effectively characterize the style consistency of the generated set in terms of overall distribution proximity and local transition plausibility. Further analysis indicates that a medium-scale training set (approximately 3,000–12,000 samples) achieves the best balance between distribution consistency and transition coverage, whereas excessive augmentation may lead to distribution calibration drift. Overall, the study demonstrates that data augmentation has a positive but non-monotonic effect on style consistency, highlighting the need for carefully designed augmentation strategies in highly constrained symbolic music generation tasks.

Keywords: Symbolic Music Generation, Data Augmentation, Style-Aware Generation, YNote Representation, GPT-2 Fine-Tuning

1. INTRODUCTION

In recent years, symbolic music modeling has made significant progress with the rapid development of deep learning technologies, enabling models to generate musically expressive works with complex structures [1]. However, compared to mainstream perceptual tasks such as images, videos, and speech, high-quality symbolic music datasets with detailed style annotations remain relatively scarce. When dataset sizes are limited, deep models are prone to overfitting or to drifting during generation toward more common “generic musical patterns,” resulting in outputs that violate essential stylistic constraints. This issue is particularly pronounced in niche and highly stylized musical traditions. When relying solely on data-driven learning, it is often difficult to simultaneously maintain generation quality and stylistic fidelity under data-scarce conditions [2].

Taking Jiangnan music as an example, previous studies have conducted systematic feature mining of Jiangnan music, constructed symbolic music databases and summarized formalizable stylistic characteristics across dimensions such as scales, modes, rhythm, phrase structure, and ornamentation [3]. The primary goal of these studies has been to understand and describe the stylistic characteristics of Jiangnan music, as well as to support AI composition workflows centered on rule-based or probabilistic models. In addition to research focusing on Jiangnan music, computational approaches have also been applied to the broader problem of cultural style representation and classification in music. Yusoff and Yasushi proposed an automatic metadata generation method for culture-based music impressions, demonstrating how stylistic and perceptual characteristics can be formalized into structured computational descriptors [4]. Similarly, Mao and Myint conducted a comparative study on multi-class SVM classifiers for music cultural style tagging, showing that machine learning models can effectively distinguish between different cultural musical styles based on extracted symbolic features [5]. Although these

studies primarily address classification and metadata generation rather than generative modeling, they highlight the importance of transforming culturally grounded stylistic knowledge into computationally operational forms, which provides conceptual support for integrating formalized stylistic constraints into data augmentation and generation frameworks. How to systematically incorporate such formalized knowledge into deep learning-based data augmentation has not yet been sufficiently discussed, particularly in the context of small-scale datasets and large language model (LLM) fine-tuning.

Data augmentation (DA) has been demonstrated to be a standard and effective approach for alleviating data scarcity. From automated augmentation policy search methods in the image domain (such as AutoAugment [6] and RandAugment [7]) to SpecAugment in speech recognition [8], DA techniques have been shown to significantly improve model generalization and robustness. However, data augmentation for symbolic music poses unique challenges. Unlike audio-level operations such as pitch shifting or time stretching, symbolic-level augmentation must simultaneously preserve musical grammatical validity and cultural stylistic fidelity [2]. For non-Western musical traditions with strict modal and cadential conventions, simple transposition or random transformations may disrupt the core stylistic context—a concern that has been explicitly identified in prior studies of Persian traditional music [9].

Based on the above motivations, this paper focuses on how formally defined stylistic knowledge can be systematically transformed into symbolic music data augmentation methods that are usable by deep learning models. Taking Jiangnan music as a case study, we propose a style-aware data augmentation framework whose main contributions are summarized as follows: (1) Drawing on the intertextual variation practices commonly observed in Jiangnan music, we design a style-aware melodic variation strategy that generates variant melodic samples through operations such as localized pitch reorganization and displacement. (2) We propose a bar-level data augmentation mechanism that integrates existing rhythmic pattern inventories with first- and second-order (2D/3D) pitch transition statistics as stylistic constraints, enabling the generation of musically coherent and controllable melodic variants. (3) We implement a set of statistical and rule-based constraint criteria that reflect stylistic characteristics of Jiangnan music—such as cadential tone distributions, phrase-final note prolongation, and conventional rhythmic patterns—to serve as generation constraints during data augmentation, thereby preventing the production of stylistically inconsistent melodic samples. (4) We propose a quantitative evaluation protocol that jointly assesses generation quality and stylistic fidelity. The proposed data augmentation framework is suitable for learning scenarios characterized by data scarcity and strict requirements for stylistic consistency, and it can also serve as a supplementary data source when fine-tuning large generative models in specific stylistic domains.

In this context, the application of large language models (LLMs) enables generative models to capture, to some extent, long-range semantic dependencies and macro-level structures, while supporting more complex stylistic transformations; consequently, LLMs are often employed to assist data augmentation. Examples include using LLMs to generate synthetic lyrics to expand training corpora [10] or leveraging language models for semantic-level augmentation in music-text alignment tasks [11]. However, LLMs typically exhibit a strong dependence on the scale of training data, and when fine-tuned on small-scale datasets, they are prone to issues such as overfitting or stylistic drift. Against this backdrop, this study aims to investigate the effectiveness of style-aware data augmentation strategies for LLM fine-tuning on small, style-specific datasets, with the goal of providing a more robust data foundation for music education, cultural preservation, and personalized creative applications.

2. BACKGROUND AND RELATED WORK

2.1 Symbolic Music Modeling

Symbolic music generation typically represents musical scores as sequences of discrete tokens and learns conditional probability distributions in a language-modeling paradigm, generating notes, durations, and structural markers in an autoregressive manner. Early studies primarily employed recurrent neural networks (RNNs) or long short-term memory networks (LSTMs) to model melodic and rhythmic structures, such as C-RNN-GAN [12], which combines recurrent neural networks with adversarial learning. Subsequently, variational autoencoders (VAEs) were introduced into symbolic music

generation to learn continuous latent musical representations and enhance generative diversity, as exemplified by MusicVAE [13]. In recent years, Transformer architectures, owing to their strong capacity for modeling long-range dependencies and their scalability, have gradually become one of the dominant research paradigms in symbolic music generation. Music Transformer improves long-sequence modeling by introducing relative positional encoding [14], while Pop Music Transformer builds upon this approach by adopting and extending the beat-based REMI representation, thereby enhancing control over rhythmic structure and expressive performance attributes [15].

Despite these advances, sequence-based models continue to face two major challenges. The first lies in modeling long-term musical structure: macro-level sections and syntactic organization are difficult to learn solely from token-level statistical relationships. The second challenge concerns the strong coupling among multiple musical attributes, which can cause models to exhibit stylistic drift during the generation process. To mitigate these issues, the Compound Word Transformer integrates multiple musical attributes into a single compound token, reducing alignment errors and computational overhead caused by excessively long sequences. However, there remains considerable room for improvement in highly constrained regional or instrument-specific stylistic scenarios [16].

2.2 Data Scarcity and High Constraint Density

Compared to mainstream Western datasets or large-scale MIDI corpora, regional and instrument-specific musical styles are characterized by small dataset sizes, dispersed sources, and inconsistent annotation formats. Under such data limitations, deep models often exhibit two typical phenomena: first, the generated outputs tend to concentrate excessively on a few high-frequency fragments to maintain local consistency; second, attempts to increase diversity may disrupt existing grammatical or structural constraints, resulting in weakened stylistic characteristics or producing implausible outcomes at phrase terminations and structural boundaries. In small-sample music generation, the absence of structural or domain knowledge often leads models to errors in maintaining overall structural coherence. Even with modern generative architectures, accurately modeling long-range thematic development and global structure remains an unresolved challenge [17]. Therefore, under conditions of limited data, simultaneously maintaining generative diversity and structural consistency remains a critical challenge in symbolic music modeling for regional styles.

2.3 Music Data Augmentation

Data augmentation has been relatively well established in the audio domain; however, in the symbolic music domain, augmentation operations must simultaneously preserve playability and stylistic plausibility. MusicBERT adopts a masked pre-training strategy as the foundation for representation learning, enhancing model robustness to missing or perturbed musical events [18]. Nevertheless, for highly constrained regional or instrument-specific styles, simple transposition or random masking may violate style-specific modal, rhythmic, or cadential rules. In such scenarios, data augmentation strategies often need to incorporate explicit stylistic knowledge—such as rule-based transposition constraints or rhythmic micro-adjustments—to ensure that the augmented sequences remain within a valid stylistic space.

2.4 Computable Representation and Markov Statistics: Foundations for Regularized Augmentation

Computable symbolic music representations allow researchers to directly manipulate attributes such as pitch, duration, and register, transforming musical scores into sequence data that can be statistically analyzed, substituted, and resampled. In this study, we adopt the YNote notation system, which encodes pitch and duration into discrete tokens with a fixed format [19], facilitating subsequent statistical analysis and data operations.

In melodic modeling and style construction, Markov chains have long served as a classical statistical baseline. Their state transition probabilities directly reflect conditional relationships among musical events in the corpus, providing a high degree of interpretability in melodic modeling. In recent years, higher-order contextual information has increasingly been incorporated into such statistics. These approaches are particularly well suited to serve as the core of style-oriented data

augmentation: under a fixed rhythmic framework, transition statistics are used to regulate local melodic plausibility, while rule-based constraints govern cadential behavior and phrase-final patterns, thereby providing more stable training signals for deep learning models [3].

2.5 Evaluation for Style: Limitations of n-gram Overlap and a Distributional Consistency Perspective

Evaluation of symbolic music generation often borrows metrics from natural language generation, such as BLEU or ROUGE, which measure the similarity between generated sequences and reference sequences based on n-gram overlap. However, prior studies have pointed out that such metrics rely heavily on surface-level overlap and struggle to accurately reflect creativity and stylistic consistency in music generation [20]. Within a given style, the number of musically valid melodic variants is large, and a low n-gram overlap score does not necessarily indicate poor generation quality.

As a result, recent studies have increasingly adopted distributional consistency metrics, such as comparing pitch histograms or duration distributions using Kullback–Leibler (KL) divergence [21]. In this context, for style-oriented generation tasks, local transition-based statistics (e.g., bigram coverage) can be used to assess whether generated outputs are consistent with the corpus statistics in terms of local continuation patterns. Based on these considerations, this study treats BLEU as an auxiliary metric and adopts distributional consistency and local transition statistics as the primary quantitative evaluation criteria.

3. METHODOLOGY

This study proposes a symbolic music modeling pipeline centered on the integration of rule-based data augmentation and language model fine-tuning, targeting training, generation, and evaluation on YNote-based representations of Jiangnan-style music. The overall workflow consists of four main stages: (1) Generating stylistically consistent augmented sequences through rule-based random search combined with Markov-style statistical modeling of musical style; (2) Fine-tuning GPT-2 using causal language modeling (CLM); (3) Generating new sequences using a sampling strategy informed by bigram statistics, followed by post-normalization; (4) Conducting quantitative evaluation using matrix-based similarity comparisons and distributional consistency metrics.

3.1 Symbolic Representation and Post-normalization

In this study, YNote is adopted as a computable notation, in which each token consists of a fixed-length string of four characters that encode pitch and duration information. This representation enables musical scores to be processed and statistically analyzed as discrete symbolic sequences. Since the raw outputs (RAW) generated by the language model may contain non-standard characters or incomplete fragments, we employ a sliding-window scanning mechanism to extract valid tokens that conform to the YNote specification and convert them into normalized sequences (NORM) for subsequent training and evaluation.



Figure 1. Overview of YNote Format Photograph by Lu et al. [19]

3.2 Rule-Based Data Augmentation: GenNewSong

To effectively increase the number of training samples under limited data conditions while preserving both local melodic continuity and global stylistic characteristics of Jiangnan music, this study adopts GenNewSong, a rule-based data augmentation procedure proposed in this work. GenNewSong is not an independent melody generation model; instead, it operates on existing melodies and performs controlled local modifications under explicit structural rules and statistical style constraints to produce stylistically consistent melodic variants, which are then used as supplementary data for subsequent deep model training.

At the structural level, GenNewSong takes the bar as the basic unit of operation. The rhythmic pattern and note positions within each bar are kept fixed, and the onset and termination note positions of each phrase are preserved. This design ensures that the augmented melodies remain consistent with the original melodies in terms of rhythmic framework and syntactic structure. By doing so, the augmentation process avoids disrupting rhythmic conventions and cadential characteristics that are common in Jiangnan music, allowing the melodic variants to retain a recognizable stylistic contour.

At the pitch level, the system computes the occurrence frequencies of adjacent note pairs from the original training corpus to construct a first-order Markov-style pitch transition model that captures commonly observed intervallic motion directions in the style. In addition, the system computes the co-occurrence frequencies of three-note sequences, forming fixed-length (trigram) pitch transition patterns that capture frequently occurring local melodic fragments. These statistical models are not used for note-by-note melody generation; rather, they serve as probabilistic constraints during candidate pitch selection, reducing the likelihood of producing unnatural pitch transitions.

During the actual augmentation process, the system performs randomized local pitch modifications on the original melody under fixed rhythmic and bar-level structural constraints. For each modifiable note position, candidate pitches that conform to stylistic statistics are generated based on the preceding pitch context, by referencing both first-order and trigram pitch transition statistics. Selection among these candidates is conducted in a probability-weighted manner rather than through unconstrained random substitution. This mechanism allows the generated melodies to maintain reasonable pitch continuity and local melodic coherence while still introducing diversity. All generated melodic variants are constrained by explicit stylistic rules throughout the generation process, including rhythmic pattern validity, phrase-initial and phrase-final pitch constraints, and pitch transitions guided by fundamental style statistics. Only melodic sequences that satisfy all of the above constraints are retained, achieving a balance between melodic diversity and stylistic stability. The validated melodies are then output in YNote symbolic representation, forming augmented training samples for subsequent model training and serving as training data for GPT-2 fine-tuning.

To enhance interpretability and facilitate human inspection, this study also provides an interactive interface that uses the same YNote symbolic representation as model training. The interface allows researchers to inspect differences between original and augmented melodies. Figure 2 illustrates the interface, where users can select source phrases from the annotated corpus, generate stylistically constrained variants, and compare original and generated phrases through auditory playback, complemented by pitch-contour visualizations for qualitative inspection. For improved readability by domain experts, the system can optionally render standard staff notation; however, this notation is provided solely for auxiliary visualization and is not involved in the model training or evaluation pipeline.

For clarity and reproducibility, the complete generation procedure of GenNewSong is formally summarized in Figure 3. The algorithm details the bar-level iterative generation process, the stochastic pitch sampling based on the transition matrix (M), and the cadential validation procedure under structural and stylistic constraints. From a musical validity perspective, the explicit validation step serves as a constructive constraint-satisfaction mechanism. By strictly enforcing that any generated sequence failing to match the designated phrase-final pitch P_{rear} is discarded and regenerated, the algorithm guarantees, by construction, that the phrase-final constraint violation rate is zero. Consequently, all retained augmented samples inherently preserve the structural integrity of the Jiangnan style without requiring any post-hoc filtering.

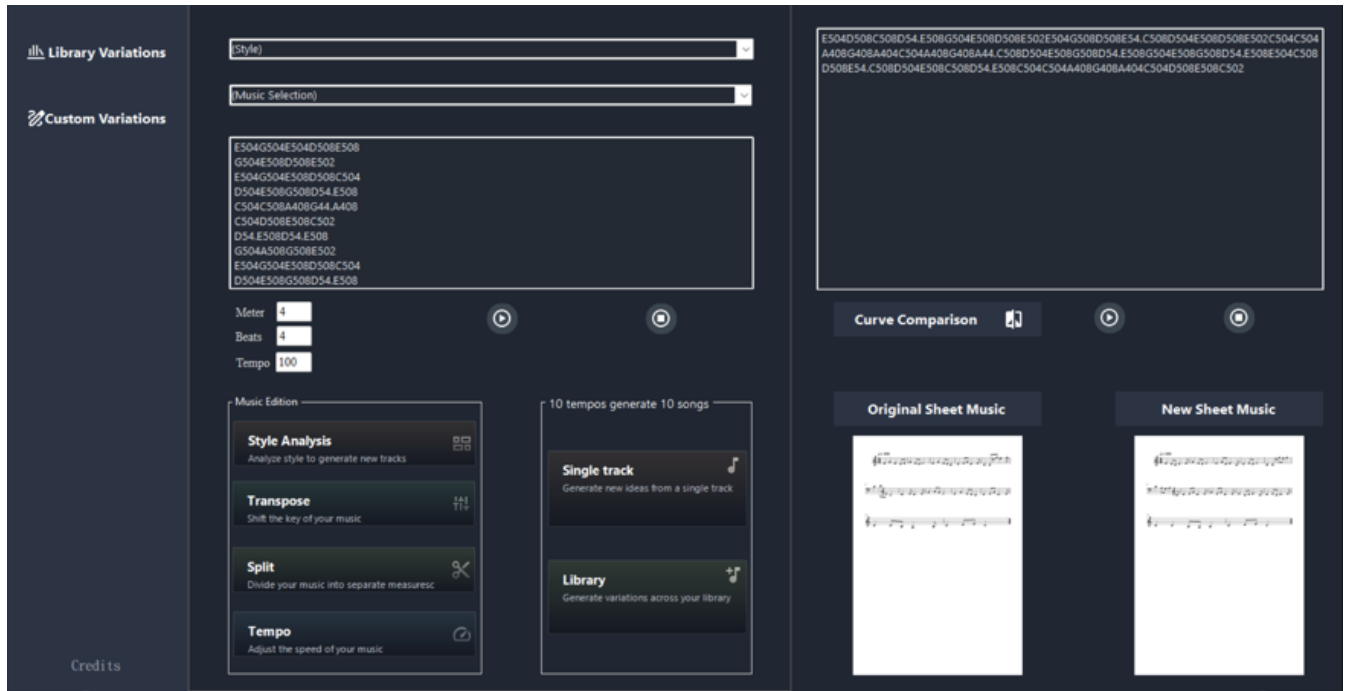


Figure 2. Interactive interface

Input:

- Original melody segmented into bars $B = \{b_1, b_2, \dots, b_N\}$
- Fixed rhythmic patterns $R = \{r_1, r_2, \dots, r_N\}$
- Head pitches $P_{head} = \{p_1^{head}, p_2^{head}, \dots, p_N^{head}\}$ and rear pitches $P_{rear} = \{p_1^{rear}, p_2^{rear}, \dots, p_N^{rear}\}$
- Pitch transition matrix M

Output:

- Augmented melody S in YNote format

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Procedure:
1: For each bar  $b_i$  in B do
2:   Repeat
3:     Initialize pitch sequence  $seq \leftarrow [p_i^{head}]$ 
4:     Target termination pitch  $\leftarrow p_i^{rear}$ 
5:     While  $\text{count}(seq) < \text{count}(r_i)$  do
6:        $current_{pitch} \leftarrow$  last element of  $seq$ 
7:        $next_{pitch} \leftarrow$  Get from  $M$  based on  $current_{pitch}$ 
8:       Append  $next_{pitch}$  to  $seq$ 
9:     End While
10:   Until last pitch of  $seq = p_i^{rear}$ 
11:   Interleave  $seq$  with  $r_i$  to form YNote tokens
12: End for
13: Return Concatenated sequence  $S$ 

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Figure 3. The pseudocode of GenNewSong

3.3 GPT-2 Fine-Tuning (Causal Language Modeling)

In the deep learning stage, GPT-2 [22] is fine-tuned to learn the conditional probability distribution of YNote sequences under a causal language modeling (CLM) framework. The training data consists of the augmented sequences generated in Section 3.2. The training set size N_{train} varies according to experimental settings, ranging from 150 to 15,000 samples, while the test set is fixed at 39 original compositions. Each sample is provided to the model as a sequence of whitespace-separated tokens.

During tokenization, the maximum sequence length is set to $MAX_LEN = 256$, and sequences exceeding this length are truncated. Fine-tuning hyperparameters are fixed as summarized in Table 1. To ensure reproducibility, the random seed is set to $SEED = 42$. If GPU resources are available, FP16 (half-precision) training is enabled to improve computational efficiency.

Table 1. Fine-tuning hyperparameters for GPT-2

Parameter	Value	Description
EPOCHS	10	Number of training epochs
LR	5e-5	Learning rate
BATCH_SIZE	2	Per-device batch size
GRAD_ACCUM	8	Gradient accumulation steps (effective batch size = 16)
MAX_LEN	256	Maximum token sequence length
SEED	42	Fixed random seed for reproducibility
FP16	Enabled if GPU available	Half-precision training for efficiency

3.4 Generation Strategy: Bigram-Aware Sampling and Post-normalization

After completing model fine-tuning, this study adopts sampling-based decoding during the generation stage to avoid the output homogenization commonly caused by deterministic decoding strategies. During generation, the model is prompted with randomly selected fragments from the training sequences and produces multiple candidate melodic sequences.

To further reduce implausible local note transitions in the generated outputs, we introduce a bigram-aware logits processor during sampling. This mechanism leverages the statistical information of adjacent token pairs observed in the training data to impose a stylistically consistent bias on the sampling distribution. In addition, to ensure that the generated outputs conform to the grammatical constraints of the YNote representation, the raw model outputs are subjected to post-normalization, which converts them into standardized token sequences for subsequent analysis and evaluation. Table 2 summarizes the parameters used during the generation stage, which can be categorized into three groups according to their functionality: prompt and output length settings, sampling-based decoding control parameters, and bigram-aware bias

parameters. The first two groups control the length and randomness of the generated sequences, while the bigram-aware parameters introduce style-consistent local transition biases during sampling, thereby reducing the likelihood of implausible continuations.

Table 2. Generation and Decoding Parameter Settings

Category	Parameter	Value	Description
Prompt & Length	PROMPT_TOKENS	30	Length of the prompt sequence
Prompt & Length	MAX_NEW_TOKENS	128	Maximum number of generated tokens
Prompt & Length	N_GEN	600	Number of generated sequences
Sampling	TEMPERATURE	0.9	Probability distribution smoothing
Sampling	TOP_P	0.95	Nucleus sampling threshold
Sampling	REPETITION_PENALTY	1.05	Penalty for repeated fragments
Bigram-aware	BIGRAM_ALPHA	0.75	Unigram–bigram mixing coefficient
Bigram-aware	BIGRAM_BLOCK_STRENGTH	9.0	Strength of bigram-aware bias
Output	RAW	-	Raw decoded output
Output	NORM	-	Post-normalized output

3.5 Evaluation Metrics

This study evaluates symbolic music generation on Jiangnan-style YNote sequences. Owing to the inherently high multimodality of symbolic music, where a single style admits a large number of musically valid variants, and the fact that stylistic characteristics are primarily manifested through distributional tendencies and local continuation vocabularies, we design two categories of evaluation metrics:

1. Reference Similarity:

To quantify the similarity between generated sequences and test pieces at the n -gram fragment level, this study adopts the BLEU family of metrics as the baseline similarity measure. Given a generated sequence h and a reference sequence r , BLEU is defined in terms of clipped n -gram precision and a brevity penalty as follows:

$$p_n = \frac{\sum_{g \in \text{grams}_n(h)} \min(\text{count}_h(g), \text{count}_r(g))}{\sum_{g \in \text{grams}_n(h)} \text{count}_h(g)} \quad (1)$$

$$BP = \begin{cases} 1, & c > r \\ \exp\left(1 - \frac{r}{c}\right), & c \leq r \end{cases} \quad (2)$$

$$BLEU = BP \cdot \exp\left(\sum_{n=1}^N w_n \log p_n\right) \quad (3)$$

where c denotes the length of the generated sequence and w_n represents the weighting coefficient for each n -gram order. This study reports BLEU-1 and BLEU-4, and additionally includes Senior BLEU, a stricter variant without smoothing, as a complementary metric. Due to the high degree of multimodality inherent in symbolic music generation, for each test sequence r_i , the model randomly generates K candidate sequences $\{g_j\}_{j=1}^K$. BLEU scores are computed for each pair (r_i, g_j) , forming an $N \times K$ score matrix S where $S_{ij} = M(r_i, g_j)$. To aggregate similarity scores under the multi-candidate generation setting, this study adopts Top-1 Average as the aggregation strategy. Specifically, for each test sequence, only the highest BLEU score among its candidates is retained, and the final score is obtained by averaging over the entire test set:

$$\text{Top-1 Avg}(M) = \frac{1}{N} \sum_{i=1}^N \max_{1 \leq j \leq K} M(r_i, g_j) \quad (4)$$

This aggregation scheme evaluates whether, under a fixed generation budget, the model is capable of producing at least one candidate that exhibits high local structural similarity to the reference, making it particularly suitable for symbolic music generation tasks with high multimodality.

2. Style Statistical Consistency

This category assesses whether the generated set preserves the global distributions and local continuation patterns of the training corpus, using the following metrics:

- (1) KL Divergence: KL divergence measures the discrepancy between the generated distribution Q and the training distribution P , where x denotes an event or symbol category and ϵ is a small constant added to avoid numerical instability. Three granularities are considered in this study: (1) KL_token: distribution over complete YNote tokens; (2) KL_pitch: pitch/register distribution derived from tokens; (3) KL_dur: duration distribution derived from tokens.

$$D_{KL}(P \parallel Q) = \sum_x P(x) \log \frac{P(x) + \epsilon}{Q(x) + \epsilon} \quad (5)$$

- (2) Bigram-based Metrics: These metrics evaluate whether local continuation vocabularies are consistent with the training corpus: (1) Bigram_seen measures the proportion of generated bigrams that have been observed in the training data; (2) Bigram_KL computes the KL divergence between the bigram distributions of the training data P_{bi} and the generated data Q_{bi} .

$$\text{Bigram_seen} = \frac{\sum_{(a,b) \in B_{train}} \text{count}_G(a,b)}{\sum_{(a,b)} \text{count}_G(a,b)} \quad (6)$$

$$\text{Bigram_KL} = D_{KL}(P_{bi} \parallel Q_{bi}) = \sum_{(a,b)} P_{bi}(a,b) \log \frac{P_{bi}(a,b) + \epsilon}{Q_{bi}(a,b) + \epsilon} \quad (7)$$

In summary, the proposed evaluation protocol jointly considers local alignment via n-gram similarity and global statistical consistency, thereby avoiding the limitations of relying on a single metric and providing a more faithful assessment of stylistic fidelity in symbolic music generation.

4. EVALUATION AND RESULTS

4.1 BLEU-based Reference Similarity

Table 3. BLEU Results (Top-1 Average; test = 39, generated = 600)

Training scale	BLEU-1 (NLTK)	BLEU-4 (NLTK)	Senior BLEU-1	Senior BLEU-4
150	0.3027	0.1122	0.3027	0.1122
1,500	0.2441	0.0444	0.2441	0.0291
3,000	0.2374	0.0410	0.2374	0.0237
6,000	0.2402	0.0421	0.2402	0.0269
9,000	0.2417	0.0419	0.2417	0.0241
12,000	0.2399	0.0424	0.2399	0.0276
15,000	0.2408	0.0408	0.2408	0.0248

Table 3 summarizes the Top-1 Average BLEU results under different training set sizes. As the amount of training data increases from 1,500 to 15,000 samples, both BLEU-1 and BLEU-4 remain within a relatively narrow range and do not exhibit a monotonic improvement with increasing data scale.

This observation is consistent with the interpretation of the Top-1 Average aggregation. Under the current evaluation setting, BLEU-based metrics primarily assess whether, given a fixed generation budget ($K = 600$), the model is able to generate at least one candidate whose n-gram structure exhibits high overlap with a reference sequence. As a result, BLEU scores are more sensitive to the concentration of the generation distribution and the repetition rate of high-frequency fragments, rather than reflecting overall stylistic consistency or stability.

Consequently, in this study, BLEU is treated as an auxiliary indicator of reference similarity, rather than as the primary metric for evaluating the effectiveness of the proposed data augmentation strategy.

4.2 KL and Bigram-based Style Statistical Consistency

Stylistic consistency evaluates whether the generated set preserves the statistical characteristics of the training corpus in terms of symbol usage proportions and local continuation patterns.

Table 4. Distributional Consistency and Local Transition Metrics (test = 39, generated = 600)

Training scale	KL_token	KL_pitch	KL_dur	Bigram_seen	Bigram_KL
150	0.2258	0.1833	0.0665	0.7601	1.7498
1,500	0.0904	0.0757	0.0003	0.9664	0.2087
3,000	0.0663	0.0499	0.0013	0.9814	0.1689
6,000	0.0742	0.0536	0.0018	0.9884	0.1697
9,000	0.1120	0.0717	0.0100	0.9923	0.2383
12,000	0.0754	0.0552	0.0028	0.9876	0.1727
15,000	0.1398	0.0919	0.0151	0.9758	0.2855

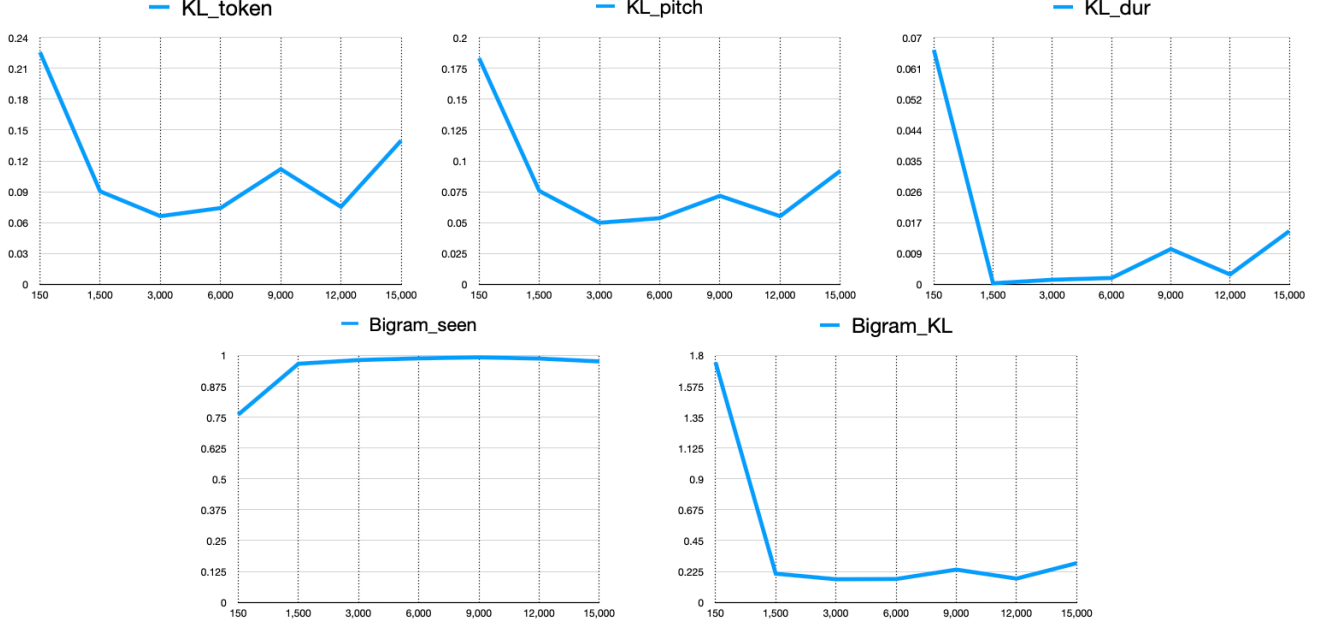


Figure 4. Trends of KL_token, KL_pitch, KL_dur, Bigram_seen, and Bigram_KL across different training set sizes

KL Divergence. As the training set size increases from 150 to the range of 3,000–12,000 samples, both KL_token and KL_pitch decrease substantially, reaching minimum values of approximately 0.0663 and 0.0499, respectively. This indicates that the generated sets become increasingly aligned with the training corpus in terms of overall token usage and pitch distribution. Meanwhile, KL_dur remains extremely low across the range of 1,500–12,000 samples, suggesting that the duration distribution is almost perfectly matched to that of the training data. However, when the training set size is further expanded to 15,000 samples, KL_token and KL_pitch increase again. This trend suggests that excessive data augmentation may introduce distributional calibration shifts, rather than continuing to improve distributional alignment.

Bigram Coverage vs. Calibration. Bigram_seen remains consistently high in the range of 6,000–12,000 samples (0.9884, 0.9923, and 0.9876), indicating that nearly all adjacent token pairs in the generated sequences fall within the set of transitions observed in the training corpus. This reflects a high degree of local continuity and stability in the generated outputs. In contrast, Bigram_KL exhibits a different pattern. The lowest values are observed at training sizes of 3,000 and 6,000 samples (0.1689 and 0.1697), whereas a clear increase is observed at 15,000 samples (0.2855). These results suggest that although larger-scale data augmentation further improves transition coverage, the relative frequency distribution of those transitions becomes increasingly misaligned with the original corpus. This highlights a trade-off between coverage and calibration in local continuation statistics.

5. CONCLUSION

By synthesizing the results in Table 3 and Table 4, it can be observed that BLEU-based metrics and KL/Bigram-based metrics reflect different levels and semantic aspects of the generation behavior. Under the experimental settings of this study, BLEU-Top1Avg primarily measures whether, given a fixed generation budget, the model can produce at least one candidate sequence that highly overlaps with a reference sequence at the n-gram level. Therefore, BLEU-Top1Avg do not exhibit a

monotonic trend with respect to training data size and are more easily influenced by the concentration of the generation distribution and the repetition of high-frequency segments.

In contrast, KL divergence and bigram statistics directly characterize whether the generated set aligns with the training data in terms of overall token distribution and local transition patterns, providing a more direct reflection of the concept of “style consistency.” The experimental results show that as the training data scale increases from small to medium (approximately 3,000–12,000 samples), the generated outputs gradually approach the training data in terms of token, pitch, and duration distributions, while the local transitions almost entirely fall within the bigram patterns observed in the corpus. This indicates that data augmentation effectively reduces style drift and enhances the local melodic plausibility.

However, further increasing the training data scale to 15,000 does not continue to improve all metrics. Although Bigram_{seen} remains at a high level, indicating continued coverage of local transitions, KL_{token} and Bigram_{KL} increase simultaneously, showing that the generated outputs deviate from the original corpus in both transition proportions and overall distribution shapes. This phenomenon suggests a trade-off between improving “coverage” and maintaining “calibration”: excessive augmentation may amplify the inherent statistical biases of the augmentation process, thereby degrading distribution calibration.

Overall, the results of this study indicate that the impact of data augmentation on style consistency exhibits a non-monotonic trend. A medium-scale training dataset achieves the best balance between distribution proximity and local transition stability, whereas too small a dataset fails to constrain style statistics, and too large a dataset may introduce proportional shifts and calibration risks. This observation highlights that in highly constrained symbolic music generation tasks, data augmentation is not merely about “increasing quantity,” but also requires careful consideration of its effects on the shape of the generated distribution and the usage proportions of local musical vocabulary.

6. LIMITATIONS AND FUTURE DIRECTIONS

First, this study adopts a data augmentation strategy that combines rule-based modifications with Markov-style statistical constraints, aiming to expand the range of observable melodic transitions within a fixed rhythmic framework while preserving the syntactic plausibility of Jiangnan-style phrases. It should be noted that, although the augmented sequences are not verbatim copies of original pieces, they remain governed by predefined transition statistics and structural rules, which may induce a certain degree of correlation among samples. Experimental results indicate that increasing the scale of augmented data improves overall alignment with the training corpus in terms of KL divergence and bigram statistics, while consistently enhancing local transition coverage. This suggests that data augmentation effectively reinforces style-consistent statistical patterns. However, when the augmentation scale becomes large, inherent biases in the augmentation process, such as repeated use of specific rhythmic templates or local substitution mechanisms, may alter the relative proportions of certain tokens or bigrams, leading to cases where coverage continues to increase but calibration begins to deviate. This phenomenon is reflected in observations where some distributional consistency metrics do not further improve under high-coverage conditions. Future work could incorporate distribution-aware reweighting or constrained sampling mechanisms during augmentation to expand coverage while more strictly maintaining frequency structures consistent with the original corpus.

Moreover, this study fine-tunes GPT-2 under varying training data scales and evaluates the effect of data augmentation on the generated distribution within a fixed generation budget and a consistent matrix-based evaluation framework. While this design facilitates comparisons across different training scales, results may still be influenced by interactions between training randomness and configuration, such as training steps, token coverage, or random seed variations. Although we attempted to control for model architecture and major training procedures to maintain comparability, future studies could employ stricter control measures, such as fixing update steps or token budgets, and report averages and variances across multiple random repetitions to improve statistical reliability and reproducibility.

In addition, this study employs complementary evaluation metrics, including reference-overlap measures (BLEU) and distributional consistency indicators (KL divergence, Bigram_{seen}, and Bigram_{KL}), to reflect different semantic levels in

symbolic music generation. Given the high degree of multi-solutions within the Jiangnan style, a single reference cannot capture all plausible variations. BLEU primarily measure the similarity of generated sequences to specific reference fragments rather than directly quantifying stylistic consistency. Furthermore, when applying Top-1 Average aggregation, these metrics are sensitive to distribution concentration and high-frequency fragment ratios and thus should not be solely relied upon for assessing style quality. By contrast, KL divergence and bigram statistics provide direct insights into the alignment of token, pitch, duration, and local transition distributions, supporting more reliable quantification of stylistic consistency. Future work may develop evaluation metrics more closely aligned with Jiangnan style, such as cadence distributions, phrase-final note patterns, or pitch constraints for terminal notes, to complement distribution-based metrics with syntactic-level assessments.

Finally, beyond pitch and rhythmic statistical tendencies, Jiangnan-style music also exhibits higher-level phrase structures and performative syntax. The current automatic evaluation focuses primarily on distributional consistency and local transition plausibility, leaving higher-level structural and performative aspects unexamined. Future research could modularize rule-based checks to perform interpretable style-violation analyses post-generation and integrate expert annotations or playability assessments, thereby establishing a multi-level evaluation framework that encompasses statistical, syntactic, and performative dimensions. Furthermore, the proposed data augmentation approach could be extended to other regional genres or instrumental styles to evaluate its generalizability and transferability in low-resource, highly constrained symbolic music generation tasks.

7. FUTURE WORK

Based on the limitations and observations discussed above, several directions for future research are envisioned. First, the evaluation framework could be extended to include explicit style-constrained metrics. Beyond distributional indicators, syntactic and phrase-final checks specific to Jiangnan style—such as terminal note pitch, phrase-final long notes, and phrase segmentation—could be incorporated, with violation rates or deviation scores serving as complementary measures to BLEU.

Second, the distributional bias inherent in rule-based augmentation could be mitigated, particularly when generating large-scale augmented datasets. This may involve reweighting or constrained sampling strategies to ensure that token and bigram frequencies more closely match the original corpus, balancing coverage with calibration.

Third, augmentation operations could be modularized and parameterized. Components such as Markov order, rhythmic templates, pitch onset/offset conditions, and local substitution rules could be made adjustable, and their individual contributions to KL divergence, Bigram_{seen}, and Bigram_{KL} could be systematically analyzed, forming a set of interpretable design guidelines.

Fourth, the proposed augmentation pipeline could be extended to other musical genres or instrumental styles, particularly those within pentatonic or regional folk traditions, to evaluate its generalizability and establish a broader framework for low-resource style-specific music generation.

Finally, evaluation could be strengthened by combining objective and subjective assessments. In addition to automatic metrics, expert annotations or playability assessments could be incorporated, potentially integrated with style-classifier models, to form a more comprehensive evaluation loop that captures both statistical and perceptual aspects of stylistic fidelity.

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